Srikar Devulapalli

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SUMMARY

Senior Data Science & AI Engineer with 5+ years of experience in Python, SQL, and R. Skilled in building scalable pipelines, predictive modeling, KPI development, Gen AI, LLMs, Agentic AI and A/B testing frameworks. Advanced in Tableau, Power BI, and BI tools. Delivered significant impact across sectors (healthcare, gaming, telecom, insurance) including $9.5M+ revenue uplift at Microsoft Teams and $5M+ fraud savings at Suncorp Insurance.

EXPERIENCE

AI Engineer — June 2025 - Present

N9 IT Solutions Inc. — Remote, USA

- Architected and deployed an AI-augmented patient cohort retrieval system using RAG + vector embeddings over EHRs, clinical notes, and biomedical ontologies — enabled natural-language cohort queries and drove a 3× increase in cohort reuse and 50% reduction in manual screening effort.
- Engineered an LLM-backed clinical query assistant indexing labs, imaging reports, and provider notes into a semantic knowledge graph + vector store — supported diagnostic QA, boosting clinician adoption by 60% and improving interdisciplinary consensus on cases by 25%.

Data & AI Engineer — October 2024 - May 2025

4i Americas LLC - Brightspeed (contract) — Remote, USA

- Architected and deployed a Retrieval-Augmented Generation (RAG) system for network operations. Built a vectorized knowledge base from IQGEO network schematics, historical outage reports, and technical manuals, enabling engineers to query complex information via natural language and reducing mean time to resolution (MTTR) by 50%.
- Engineered a robust ETL framework on GCP (BigQuery, Cloud Composer, Datafold, DataFlow), orchestrating 50+ Airflow DAGs for data warehouse objects with alerting and retry logic to automate ingestion from 15+ systems post-demerger—stabilized reporting, reduced CI/CD pipeline failures by 60%.

Data Product Manager — June 2023 - August 2023

NBC Universal — New York, NY

Driving successful delivery and governance of cloud infrastructure and Carbon Emissions Reporting tool projects.

- Led the implementation of NBC's first-ever internal emissions reporting system, setup cost-optimal cloud infrastructure to develop a carbon emission reporting tool aimed at achieving carbon neutrality. Managed requirements gathering and data ingestion processes for a unified emissions tool that identified low-hanging fruits and cutting Scope 2 emissions by 5%.
- Developed and implemented monitoring metrics for a Random Forest model predicting user re-watch behavior, optimizing targeted advertising and achieving a 30% reduction in detection time and a 25% improvement in response time.

Product Analyst — September 2021 - July 2022

Electronic Arts (EA) — Hyderabad, India

Product Analyst driving gaming engagement and monetization for Madden Mobile.

- Constructed a New User monetization funnel in Tableau and successfully advocated in reducing friction for new users to spend in-game currency, which led to a 15% increase in New Users converting to paid.
- Implemented an ARIMA model in R to forecast in-game currency spend with time series analysis, which, after UX research and A/B testing, guided a key pivot in pricing strategy, achieving an 8% increase in monthly recurring revenue.
- Created an Engagement and Monetization dashboard in Tableau that steered a revamp in enhancing product features & product placement, increasing in-app purchase volume, which led to a 12% increase in user's monthly spend.

Data Scientist (Microsoft Teams, Suncorp) — June 2018 - September 2021

Mu Sigma Inc. — Bengaluru, India

Sequoia-backed, unicorn consulting firm. Data Scientist on an engagement with the Microsoft Teams Data & Growth team. Helped scale the Teams user base, from 15M to 250M MAU. Data scientist on an engagement with HR data analytics teams at Suncorp, a leading Australian fintech.

- Developed and productized a Churn Prediction machine learning model in Python and deployed using Azure Data Factory for Microsoft Teams that informed targeted marketing efforts by A/B testing through experimental design via PowerBI report which led to a 9% increase in month-over-month retention.
- Developed and deployed ML-driven anti-abuse signals in Microsoft Teams (250M+ MAUs) that fused behavioral telemetry with graph-based anomaly detection to surface bad actors—spammers, phishers, and fraudulent account creators—cutting abuse incidents by 18% over 12 months while keeping the false-positive rate below 0.3%
- Piloted data engineering efforts to clean and standardize Microsoft Teams' calling, meeting, and device data in Databricks and built a self-serve business intelligence report suite in PowerBI for the CMD group.
- Designed a Quality of Hire dashboard for Suncorp based out of ETL framework that influenced changes to bonus structure, process improvement and staff hours policy decision-making leading a 6% increase in employee retention.
- Created a classification model in R for identifying reactivated user patterns for Microsoft Teams, which informed targeted marketing campaigns across the M365 product suite and led to a 4% increase in month-over-month reactivated users.
- Built fraud detection ML models in Python for Suncorp Insurance, saving $5M annually, and designed retention dashboards that improved policyholder retention by 7.5%.

SKILLS

Programming Languages: Python, SQL, R, Java, C, C++, Arduino, Scala, Apache Spark, MATLAB
Web Development: HTML, CSS, JavaScript, Django, React, GraphQL, AngularJS, Vue
Database Tools: Databricks, Jenkins, MongoDB, PostgreSQL, Git,Hive, Redshift, BigQuery, Cosmos DB, Snowflake, DBT
Cloud Technologies: Azure, Amazon Web Services(AWS)(EC2, S3, DynamoDB, Lambda), GCP, Docker, Kubernetes, Flink
Analytics Tools : Tableau, Power BI, Looker, Scuba Analytics, Excel, Cognos, Matplotlib, Seaborn, Plotly, SciPy
AI/ML : Statistics, Numpy, Pandas, Scikit-learn, PyTorch, TensorFlow, XGBoost, SageMaker, LangChain, HuggingFace

PROJECTS

Stock Recommendations using Stock Price Prediction and Sentiment Analysis | Python, Long Short-Term Memory, NLP
Developed a web application to provide stock recommendations that utilize a combination of stock price forecasting and sentiment analysis to assess public opinion about a company, favoring users to achieve their investment goals.

Global Football Stadiums Analysis | Python, Docker, Apache Airflow, Azure Spark Pool, Azure Synapse, NoSQL
Built an end-to-end data pipeline using Python (data extraction), Docker (deployment), Airflow (orchestration), Azure Spark (processing), and Azure Data Explorer (analytics & data mining) to uncover global stadium trends.

Resume and Job Description Analysis Tool | Streamlit, LLM, RAG, Langchain, Python, Elasticsearch, NLP
Developed an application using RAG based LLM to analyze resumes, job descriptions and implemented Chains-of-Thoughts (CoT) reasoning with GPT-3.5-turbo and Chroma for personalized, context-rich responses without model hallucinations. Hosted using Streamlit and ensured local execution with API keys.

Malaria Detection using Deep Learning | Python, CNN, Flask, Transfer Learning, PyTorch, Computer Vision
Designed an app based on Convolutional Neural Network model to automate malaria detection from blood smear images, achieving enhanced accuracy (by 6%) and robustness. This model outperformed a ResNet-152 baseline and utilized transfer learning from ImageNet and RadImageNet models, significantly reducing diagnosis errors and time.

Graph RAG Financial Crimes Investigator | Flask, Neo4j AuraDB, LangChain, OpenAI GPT-4, Vis.js, GraphQL
Built an end-to-end Graph RAG application to analyze the ICIJ Offshore Leaks dataset, integrating Flask, Neo4j, and LangChain with GPT-4 to translate natural language queries into Cypher and generate visualized insights. Automated large-scale CSV ingestion with Pandas to uncover hidden financial relationships. .

Persona Context Manager | Python, Streamlit, FastMCP, Cursor
Developed Persona Context Manager, a context-rich AI application powered by a custom MCP server, using Python, Streamlit, FastMCP, and OpenAI GPT-4, to manage and deliver structured user personas in real-time to backend LLM APIs and AI agents via a modern, editable web interface.

Vizey — GenAI Data Visualization Platform | React, Node.js, PostgreSQL, WebSockets, TypeScript, Vue, FastAPI
Developed a generative-AI application that transforms raw datasets into theme-aware visualizations via a 4-agent pipeline (Query → Data → Design → Configuration). Engineered over 80 chart types, built automatic WCAG-compliant contrast, and theme logic (fonts, gradients, textures).

Fine-Tuning a Multimodal Reward Model for Website Interactions | LLaVA 1.5, Reward Model, CLIP, RLHF
Fine-tuned a multimodal reward model using preference data to assess alignment between webpage screenshots and candidate responses. Integrated a custom scalar reward head into MM-LLM LLaVA-1.5 for downstream RLHF pipelines, enabling UI-aware reasoning. Optimized training with 4-bit quantization, gradient checkpointing, and multi-device acceleration, and built an evaluation framework to benchmark reward accuracy and layout consistency.

WorldBlend — Immersive Ad Studio | Python, JavaScript, TensorFlow.js (USE), Tailwind
Built a client-side AI ad app that analyzes video frames, ranks products via embeddings + LLM, and inserts them occlusion-aware. Includes adaptive TTS, editable catalog, and JSON/pixel verification.

Netflix-Style ELT Pipeline | dbt, Snowflake, Amazon S3
Built a modular ELT pipeline (S3 → Snowflake via dbt) with incremental models, ephemeral logic, SCD Type 2 snapshots, schema + custom tests, and auto-generated lineage docs—delivering faster refreshes.

CareTwin — AI Healthcare Digital Twin | Streamlit, Python, Pandas, Lifelines (Cox), FHIR (R4), Matplotlib
Built an AI-powered digital twin that transforms electronic health records (EHR, HL7 FHIR) into personalized risk predictions and interactive "what-if" scenarios, providing plain-language insights to support clinical decision-making.

EDUCATION

Master of Science in Data Science — **August 2022 - May 2024**
Indiana University Bloomington — Bloomington, IN - GPA 3.7/4

CERTIFICATION

AWS Certified Machine Learning Engineer - Associate — **May 2025 - May 2028**